Design of an Automated Dry-Food Dosing and Controlled Moistening System for Cats with Mastication Difficulties

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***Abstract*—Cats with reduced masticatory capacity due to periodontal disease, tooth loss, or age-related oral deterioration face mechanical difficulties ingesting dry kibble. No existing automated feeder integrates discrete volumetric dosing with on-demand moisture conditioning in a single sequence. This paper presents the design and first-prototype implementation of a mechatronic system combining a Geneva-indexed pocketed disk, a closed two-gate ultrasonic moistening chamber, and a Raspberry Pi supervisory controller in a deterministic five-state cycle that physically isolates bulk dry storage from the wet-conditioning stage. Preliminary sizing yielded a torque requirement of 0.441 N·m (safety factor: 3), met by a 10 kgf·cm servomotor with a 2.2 nominal margin. Open-loop dosing yielded an 18.3% mean error against the 12 g target (vs. 15% threshold), corresponding to ~6 kibbles without compromising device functionality or feline nutrition. Single-atomizer moistening (108–110 kHz) achieved an 8.3% relative mass gain over 30 min; three simultaneous atomizers reduced conditioning time to 5–12 min (60–83% reduction), consistent with the linear flow-rate model. These results confirm functional feasibility of the integrated sequence and establish a quantitative baseline for closed-loop dosing refinement and instrumented moistening characterization.**

***Keywords*—automatic pet feeder, mechatronic design, food dosing, food moistening, food texture conditioning.**

I. INTRODUCTION

The mechanical properties of commercial dry kibble impose a functional constraint on cats with reduced masticatory capacity: its rigid structure may increase the mechanical demands associated with food intake in animals affected by periodontal disease, tooth resorption, or age-related dental deterioration — conditions reported in more than 15% of cats under primary veterinary care, with age identified as an important risk factor for periodontal disease [1]. Although dry kibble remains the dominant format in companion-animal feeding due to its ease of storage, long shelf life, and compatibility with automated dispensing mechanisms, its mechanical properties are rarely considered as a variable to be controlled within the feeding process itself.

Automated feeding systems for companion animals provide functions such as programmable scheduling, portion control, and, in some models, remote supervision [6], [7]. However, the reviewed systems do not incorporate a downstream texture-conditioning stage for dry kibble. Wet-food feeders, by contrast, operate on pre-conditioned food and are designed primarily for preservation and controlled release rather than texture conditioning [7]. Manual moistening partially bridges this gap but introduces variability associated with the amount of water applied and the contact time, while also requiring caregiver intervention at each feeding event [3]. The reviewed systems did not identify an architecture integrating bulk dry-food storage, discrete dosing, and on-demand texture conditioning within a single automated sequence.

This paper presents the design and preliminary evaluation of a mechatronic system that closes that functional gap. The proposed architecture combines a bulk dry-food hopper, a pocketed indexing disk driven by a Geneva mechanism for discrete portion control, and a sealed ultrasonic misting chamber that conditions the kibble texture immediately before delivery. The system is designed to operate without pre-conditioned food, without refrigeration, and without caregiver intervention at each feeding event. From a robotics perspective, the system is a stationary service device that autonomously portions and conditions a granular, deformable material, a task shared with robotic food-handling and granular-manipulation systems [13], [14].

The specific contributions of this work are:

- *System architecture:* A sequential mechatronic pipeline — storage → discrete dosing → on-demand moistening → delivery — that has not been reported in the existing literature for companion-animal feeding.
- *Dosing characterization:* Experimental evaluation of the indexing mechanism under a 12 g nominal portion, with a mean absolute error of 18.3%, providing a preliminary quantitative characterization of this dispensing principle for dry kibble.
- *Moistening feasibility:* Functional validation of the ultrasonic misting stage as a means of texture conditioning, establishing baseline parameters for conditioning time and water-mass integration.

The remainder of this paper is organized as follows: Section II reviews related work and formalizes the design gap; Section III describes the mechanical and electronic architecture; Section IV presents the experimental results; and Section V states conclusions and directions for future work.

II. RELATED WORK AND DESIGN GAP

A. Automatic Dry-Food Dispensing Systems

Automated dry-food dispensers for companion animals employ three dispensing principles: gravity-gate, screw conveyance, and pocketed-disk indexing. Gravity-gate mechanisms are mechanically simple but offer no reliable portion control, as dispensed mass depends on bulk-flow conditions, hopper

geometry, and kibble morphology. Screw-conveyor systems transfer food via a motor-driven helical element and admit gravimetric feedback for portion regulation [5], [10], however, irregular kibble geometries increase jamming risk and complicate cleaning. Jain et al. [9] proposed an IoT-enabled smart pet feeder using a similar auger-based architecture integrated with remote scheduling, confirming that recent work in this domain has focused on connectivity rather than dispensing-mechanism innovation. Boateng and Akparibo [10] implemented a gravimetric-feedback dog feeder in which a load cell closes the dispensing loop, demonstrating measurable repeatability improvements over open-loop designs. Commercial gravity-feed systems such as the PETLIBRO Granary [6] illustrate the current state of the art at the product level: programmable scheduling with no mechanism for downstream texture modification.

Pocketed-disk or compartmented rotary mechanisms associate each indexing step with a discrete nominal portion, decoupling dispensed volume from bulk-flow variability and eliminating the continuous-flow risk inherent in screw-conveyor designs. Critically, across all three dispensing principles, no reported system incorporates any form of downstream texture conditioning: kibble is delivered in the same mechanical state in which it was stored [5], [6].

B. Automated Wet-Food and Refrigerated Feeding Systems

Wet-food feeders address a fundamentally different operational constraint — the spoilage of pre-conditioned food at ambient temperature — and are therefore built around an architecturally incompatible design. Systems in this category use semiconductor cooling or ice packs to maintain pre-portioned food at 4–15 °C for intervals of up to 72 hours, with sealed compartments rotated or uncovered at scheduled times [7]. Because the food is loaded already in its final soft state, no dispensing mechanism compatible with loose dry kibble is required or used, and no texture-conditioning subsystem is present. The conditioning function is performed manually by the caregiver before loading. Refrigerated wet-food feeders are therefore structurally incompatible with bulk dry-food storage and with on-demand texture modification.

C. Ultrasonic Food Conditioning

Three moistening principles are applicable to granular dry food: direct spraying, controlled immersion, and ultrasonic atomization. Direct spraying is mechanically simple but produces localized surface wetting when food remains static during application. Controlled immersion achieves deeper hydration but requires handling excess liquid and introduces a mechanical separation step before delivery.

Ultrasonic atomization generates fine droplets — typically 5–20 μm — through piezoelectric vibration of a liquid-wetted membrane, as characterized by Wang et al. [8] who modeled the relationship between driving frequency, input power, and droplet diameter in multi-modal transducers. When atomization is confined within a closed chamber, the mist distributes uniformly over the food surface, and the degree of conditioning can be regulated through exposure time alone, without requiring liquid removal. Beyond droplet generation, power ultrasound (20–100 kHz) is documented as a structural modifier of food matrices: Pau et al. [11] demonstrate that ultrasonic treatment modifies protein conformation and starch crystallinity in multiple food products; Pandiselvam et al. [12] show that ultrasonic pre-treatment accelerates moisture diffusion into porous food structures — a mechanism directly relevant to kibble rehydration. Together, these results provide a scientific basis for the selection of ultrasonic mist as the conditioning modality in the proposed system.

Manual moistening — adding water to kibble immediately before serving — partially addresses the texture requirement but introduces uncontrolled variability as a function of water volume and contact time, and is contingent on caregiver presence at each feeding event [2], [3].

D. Design Gap

The survey above reveals a structural partition in the existing design space: dry-food systems dispense without conditioning; wet-food systems preserve a pre-conditioned state without dispensing; and manual moistening depends on the caregiver at each feeding event [2], [3]. No reported system integrates bulk dry-food storage, discrete portion control, and on-demand texture conditioning within a single automated sequence. Table I formalizes this partition and positions the proposed architecture.

TABLE I
Functional Comparison of Automated Feeding Approaches

Approach	Dry-food dosing	Controlled moistening	On-demand integration
Dry-food feeder [5], [6]	Yes	No	No
Wet-food feeder [7]	No	No	No
Manual moistening [2], [3]	Manual	Partial	No
Proposed design	Yes	Yes	Yes

The contribution of the present work is therefore not an incremental improvement to either existing category, but the integration of dry-food dosing and on-demand texture conditioning into a unified mechatronic architecture, a combination unreported in the literature to date.

III. SYSTEM DESIGN

A. Design Requirements and Methodology

System requirements followed the VDI/VDE 2206 mechatronic framework [4], which structures development along concurrent mechanical, electronic, and software domains. The functional specification was derived from three primary constraints: (i) clinical — the system must reliably soften dry kibble to a texture tolerable for cats with mastication deficits; (ii) quantitative — a 12 g dosing resolution reflecting feline grazing behavior [2], with a target volumetric error of ≤ 15% to accommodate granular packing variability, kibble quantization, and anti-jamming mechanical clearances [3]; and (iii) hygienic — dry storage must be physically isolated from the moistening stage at all times to prevent condensation from propagating retrograde into the hopper. Additionally, a 4 L dry-food hopper (approx. 1.6 kg) and a 1 L water reservoir were proportionally sized based on the nominal atomization rate to sustain synchronous multi-day autonomy without partial user intervention. The resulting functional architecture is shown in Fig. 1.

B. System Architecture

The system adopts a vertical gravity-assisted layout organized into three functionally distinct stages: (1) dry-food storage and

discrete dosing, (2) closed-chamber ultrasonic moistening, and (3) supervisory electronic control. The vertical arrangement

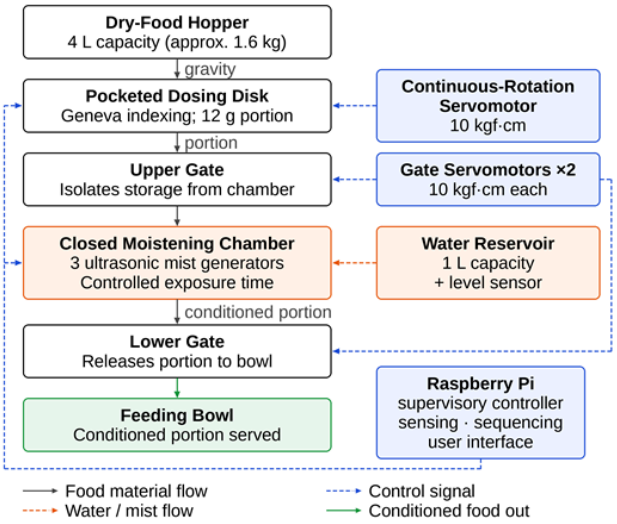


Fig. 1. Proposed automated feeding system architecture.

reduces the need for active food transfer between stages, allowing the kibble to move primarily by gravity through the upper and lower gates. The overall system has a cylindrical form factor (Ø 200 mm × 500 mm height), providing sufficient space for the hopper, dosing mechanism, and moistening chamber while maintaining a compact configuration. The CAD model and the implemented prototype are shown in Figs. 2(a) and 2(b), respectively.

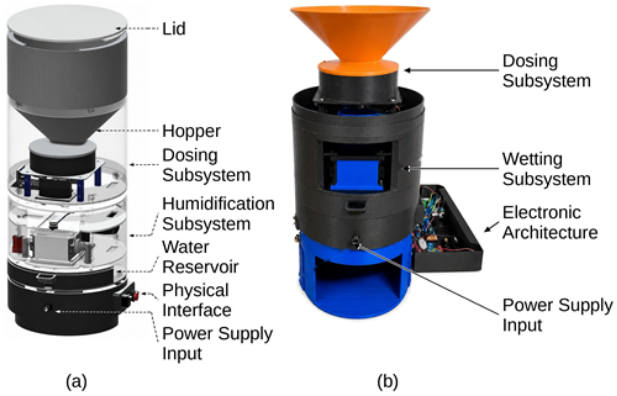


Fig. 2. (a) Complete CAD model of the proposed system; (b) implemented automated food dosing and moistening prototype.

C. Dry-Food Dosing Subsystem

Volumetric dosing utilizes a pocketed disk driven by a Geneva mechanism for discrete angular indexing, actuated by a continuous-rotation servomotor whose indexing steps are counted by a slotted infrared sensor. This architecture decouples dispensed mass from bulk-flow variability, eliminating the uncontrolled continuous-flow risk inherent to gravity-gate and screw-conveyor alternatives.

The disk has six pockets spaced at 60°, each sized to hold one nominal portion. For a nominal portion m_{nom} = 12 g and a kibble bulk density $\rho_b \approx 0.40$ g/cm³ (consistent with 1.6 kg in the 4 L hopper), the volume of each pocket is $V_p = m_{nom}/\rho_b = 30$ cm³.

While the hopper stores up to 4 L (approx. 1.6 kg) of kibble, the Janssen effect—where wall friction supports granular mass [15]—reduces the active load. Consequently, the dynamic load on the disk was conservatively modeled as a 0.5 kg reference mass at a 3 cm effective radius. As a further conservative simplification, the full weight of this mass is taken as the tangential resisting force on the disk, which corresponds to a friction coefficient of 1 and exceeds the 0.35–0.45 expected for kibble on PETG. The required torque is therefore:

$$T_{req} = m\, g\, r\, F_s \quad (1)$$

where T_{req} is the required torque (N·m), m the reference mass (kg), g = 9.81 m/s² the gravitational acceleration, r the effective radius (m), and F_s a dimensionless safety factor.

The safety factor was set to $F_s = 3$ to cover load sources that the static model omits and that were not measured: (i) uncertainty in the Janssen reduction (wall friction and lateral pressure ratio of this kibble); (ii) transient shear of kibble trapped at a pocket edge; and (iii) the inertial torque of the Geneva wheel, which accelerates from rest at each step. This value is consistent with design-factor practice under uncertain loads with moderate shock [16]. Substituting m = 0.5 kg and r = 0.03 m into (1) gives:

$$T_{req} = (0.5)(9.81)(0.03)(3) = 0.441 \text{ N·m} \approx 4.5 \text{ kgf·cm} \quad (2)$$

This requirement (≈ 4.5 kgf·cm) is met by a 10 kgf·cm continuous-rotation servomotor, which provides a margin of 2.2. The complete dosing subsystem is shown in Fig. 3.

To overcome nominal bearing friction, inertial indexing transients, and high-shear kibble quantization forces at the hopper interface, a 10 kgf·cm continuous-rotation servomotor was selected, yielding a nominal safety factor of 2.2—a selection margin over the already-factored requirement rather than an additional design factor—sufficient to absorb the inertial transients characteristic of Geneva-mechanism indexing. The complete dosing subsystem is shown in Fig. 3.

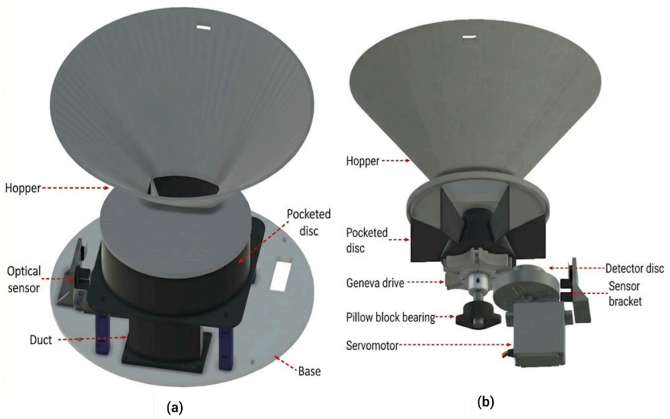


Fig. 3. CAD views of the dry-food dosing subsystem

D. Closed-Chamber Moistening Mechanism

The moistening subsystem receives the previously dosed portion and temporarily holds it inside a closed chamber bounded by an upper and a lower gate. Three ultrasonic mist generators are connected to a water reservoir to concentrate moisture around the kibble while limiting propagation toward the hopper and other components.

An initial preliminary trial indicated that a single generator required approximately 30 min to produce perceptible softening

in a 12 g portion. The preliminary sizing used a nominal atomization rate of 30–50 mL/h per module. For one generator operating for t = 0.5 h, the nominal generated mist volume is therefore:

$$V_a = Q\,t \tag{3}$$

$$V_a = (30\text{–}50\ \text{mL/h})(0.5\ \text{h}) = 15\text{–}25\ \text{mL} \tag{4}$$

This quantity is a generator-side volumetric estimate and should not be interpreted as water absorbed by the food, because part of the generated mist can remain suspended, condense on chamber surfaces, or be otherwise lost.

To reduce conditioning time, three atomizers were operated simultaneously. Assuming equivalent units and approximately additive nominal flow rates, the combined rate is 90–150 mL/h. Assuming further that the softening endpoint is reached when the same mist volume V_a has been generated, a simple linear estimate gives:

$$t_{3,\text{est}} = V_a / (3Q) = Q\,t_1 / (3Q) = t_1 / 3 \approx 10\ \text{min} \tag{5}$$

Since V_a was obtained from the same rate Q in (3), Q cancels and the estimate depends only on the single-atomizer reference time t_1 = 30 min; it therefore equals 10 min for any rate in the nominal range (15 mL / 90 mL/h = 25 mL / 150 mL/h ≈ 0.167 h). This linear estimate neglects condensation, spatial mist distribution, and possible interaction among the atomizers. Nevertheless, it was consistent with preliminary integrated-system observations, in which a qualitatively similar softening condition was reached in approximately 5–12 min. Fig. 4 shows the chamber and gate actuation mechanism.

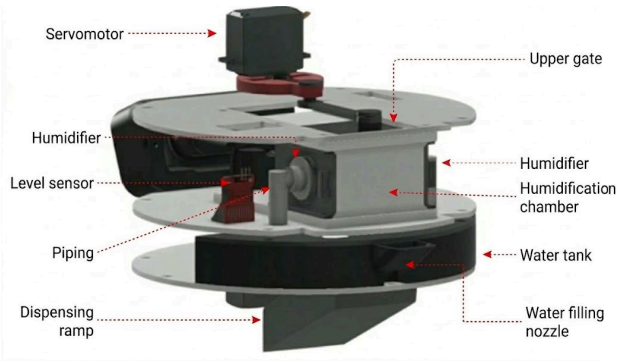


Fig. 4. CAD view of the moistening chamber and gate mechanism.

The gates are actuated independently by 10 kgf·cm servomotors and remain closed during moistening. At the end of the configured interval, the lower gate opens and releases the food toward the bowl. Water availability is supervised using a non-contact level sensor so that an insufficient-water condition can be detected before a cycle begins.

E. Electronic Architecture and Control

System supervision is implemented on a Raspberry Pi, which coordinates actuator sequencing, timing, and user communication. Sensing inputs include a hopper-level sensor and the water reservoir level sensor; both are evaluated before each feeding cycle to verify operational conditions. A USB camera facing the feeding area is also connected to the supervisor. Actuation outputs comprise the dosing servomotor, the two gate servomotors, and the three ultrasonic generators, each driven through dedicated power and switching stages. A real-time clock module enables scheduled feeding without continuous host uptime. Wi-Fi connectivity allows remote parameter configuration and status monitoring; however, the core feeding cycle executes locally and does not require a network connection. Moistening duration is controlled by a fixed time parameter in the current prototype; closed-loop humidity sensing is identified as a planned upgrade. The complete electronic architecture is presented in Fig. 5.

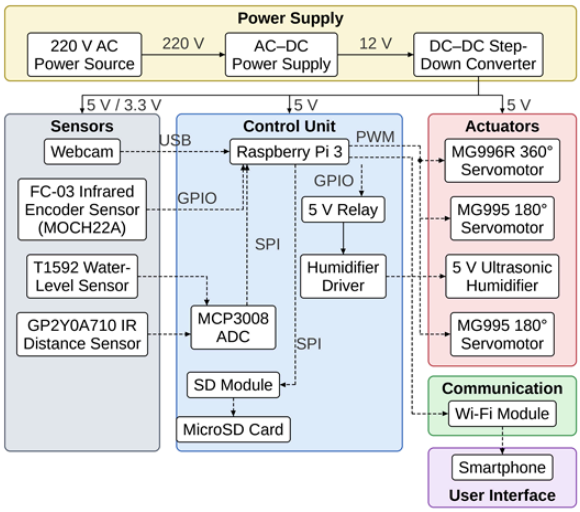


Fig. 5. Supervisory control and power architecture for the subsystems.

F. Operating Sequence

The system executes a deterministic five-state cycle — Standby → Dosing → Isolation → Moistening → Release — summarized in Fig. 7. Kibble advances through the pocketed disk until the target pocket count is discharged, after which the upper gate closes to physically isolate the dosed portion inside the chamber. The three ultrasonic generators then activate for the configured exposure interval with both gates closed, confining moisture to the conditioned portion and preventing retrograde condensation toward the hopper. Upon interval completion, the lower gate opens and delivers the portion to the feeding bowl.

Transitions in this cycle are guarded by sensor events rather than by fixed delays alone, so the supervisor operates as a discrete-event controller [17]. Table II lists, for each sensor, the decision it informs and its effect on the behavior of the system.

TABLE II
ROLE OF EACH SENSOR IN THE SUPERVISORY CONTROL

Sensor	Decision informed	Effect on behavior
Optical index sensor	Dosing → Isolation when the target pocket count is reached; jam if no pulse within 3 s	Stops dosing at the programmed count; reverses the disk 15° and retries after a jam
Hopper level sensor	Guard on Standby → Dosing	Blocks the cycle and alerts the user below threshold
Water level sensor	Guard on Standby → Dosing; checked during Moistening	Blocks the cycle; stops the atomizers and alerts the user
Camera	Presence of the cat (on-demand mode)	Starts the cycle only when the cat is detected
Software timer	Exposure time elapsed	Moistening → Release (open loop on texture)

Thus, the loop is closed at the machine level (index feedback at every step; level and timeout events start, block, or abort the cycle) but open at the process-outcome level: dispensed mass is

inferred from the pocket count and softening from exposure time. The supervisor adapts to the state of the machine and its resources, but not yet to the state of the food.

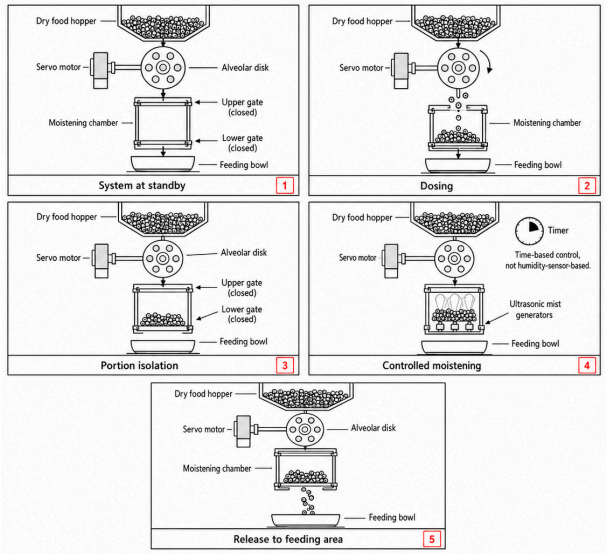


Fig. 6. Operating sequence of the automated feeding system.

IV. PRELIMINARY EXPERIMENTAL EVALUATION

A. Experimental Setup and Evaluation Scope

The experimental evaluation characterized the functional performance of the integrated prototype through independent tests of the pocketed-disk dosing mechanism and the ultrasonic moistening stage. Dosing performance was evaluated relative to a nominal 12 g portion using absolute percentage error, whereas moistening performance was assessed through relative mass gain and conditioning time for single- and three-atomizer configurations. Tests were conducted using commercial dry cat kibble under ambient laboratory conditions in Lima, Peru (18–22°C; 65–75% relative humidity). Mass measurements were obtained using a digital scale with a 1 g resolution. This resolution is coarse relative to the changes being measured: the standard uncertainty of a single reading is 1 g/√12 ≈ 0.29 g [17], which corresponds to 2.4% of a 12 g portion. A balance with 0.01 g resolution is therefore recommended for subsequent tests. All reported masses are net values, since the scale was tared with the empty moistening chamber.

B. Dosing Accuracy Evaluation

The performance of the dosing subsystem was evaluated using a nominal portion of 12 g. Ten dispensing trials were carried out under similar hopper-filling conditions. For each trial, the mechanism completed one dispensing cycle and the delivered food mass was measured immediately after discharge using a digital scale with a resolution of 1 g. Since the current prototype does not incorporate gravimetric feedback, the dosing mechanism operated in an open loop throughout the evaluation.

The absolute dosing error for each trial (i) was calculated as:

$$\varepsilon_i = |m_i - m_{nom}| / m_{nom} \times 100 \quad (6)$$

where m_i is the measured dispensed mass and m_{nom} = 12 g is the nominal target portion. Table II reports the individual trial results and aggregate error values.

TABLE II
DOSING REPEATABILITY TRIAL RESULTS (NOMINAL PORTION: 12 G)

Trial	1	2	3	4	5	6	7	8	9	10	Mean	SD
Mass (g)	14	10	15	10	14	10	14	9	14	10	12.0	2.4
Abs. error (g)	2	2	3	2	2	2	2	3	2	2	2.2	–
Error (%)	16.7	16.7	25.0	16.7	16.7	16.7	16.7	25.0	16.7	16.7	18.3	–

The experimental results yielded a mean dispensed mass of 12.0 g and a mean absolute dosing error of 18.3%, exceeding the ≤15% target established in Section III-A. The maximum and minimum absolute errors were 25.0% and 16.7%, respectively. Since the mean equals the nominal portion (95% CI: 12.0 ± 1.7 g), no systematic bias was detected; the error is due to dispersion. Masses alternate between 14–15 g (odd trials) and 9–10 g (even trials), suggesting carry-over between cycles or pocket-to-pocket differences rather than random scatter. This variability is consistent with volumetric dispensing of irregular granular materials, where differences in particle size, geometry, and packing can affect the mass transferred per dosing cycle [3], [5]. Consequently, the 18.3% mean error represents the preliminary open-loop performance of the current mechanism. Future improvements should focus on optimizing the pocket geometry and incorporating gravimetric feedback to reduce dosing variability [5].

C. Moistening Performance Evaluation

Preliminary moistening trials were conducted using 12 g portions of dry kibble. Perceptible softening was assessed qualitatively by the authors through visual inspection and manual compression of individual kibbles, and was defined as the point at which kibbles deformed under light finger pressure instead of fracturing. With a single atomizer, approximately 30 min were required to obtain perceptible softening, while the portion mass increased from approximately 12 g to 13 g. The relative mass gain was calculated as:

$$G_m = ((m_f - m_i) / m_i) \times 100 \quad (7)$$

Using m_i = 12 g and m_f = 13 g, the resulting mass gain was approximately 8.3%. Given the 1 g scale resolution, this difference carries a combined standard uncertainty of about 0.41 g (≈ 3.4 percentage points), so the value indicates that measurable water uptake occurred rather than a precise uptake figure. Fig. 8 shows the visual difference between the dry and moistened kibble.



Fig. 8. Visual comparison of kibble before and after ultrasonic moistening: (a) dry kibble and (b) moistened kibble.

With three atomizers operating simultaneously, a qualitatively similar softening condition was reached in approximately 5–12

min. Relative to the 30 min single-atomizer baseline, the conditioning-time reduction was:

$$R_t = ((t_1 - t_3) / t_1) \times 100 \approx 60\%–83.3\% \quad (8)$$

These results indicate that increasing the atomization capacity substantially reduced conditioning time. However, texture evaluation remained qualitative, and no instrumental hardness, final moisture-content, or complete water-mass-balance measurements were available. Therefore, the results should be interpreted as preliminary functional validation rather than as a clinical characterization of food softness. Objective validation will require kibble fracture force (texture analyzer) and final moisture content, with a threshold anchored to a reference such as manually moistened kibble.

V. CONCLUSIONS AND FUTURE WORK

A. Conclusions

This work presented the design and first-prototype implementation of an integrated mechatronic system for discrete dry-food dosing and on-demand ultrasonic moistening for cats with reduced masticatory capacity. The architecture combines a pocketed indexing disk driven by a Geneva mechanism, a closed two-gate moistening chamber with three ultrasonic transducers, and a Raspberry Pi controller that executes a fixed five-state feeding cycle. The vertical layout physically separates dry-storage and wet-conditioning stages throughout operation. The controller uses its sensors only to verify the state of the machine—counting disk pockets and checking food and water availability—whereas the dispensed mass and the degree of softening are not measured and are therefore controlled in open loop. Weighing each portion with a load cell and ending moistening at a measured humidity would close these loops and increase the autonomy of the system.

Preliminary sizing yielded a torque requirement of 4.5 kgf·cm (safety factor: 3), met by a 10 kgf·cm servomotor with a nominal margin of 2.2. Dosing trials produced a mean error of ~18.3% relative to the 12 g target, attributable to volumetric variability in open-loop dispensing of irregular kibble. Moistening trials established an 8.3% relative mass gain over 30 min with one atomizer; three simultaneous atomizers reduced conditioning time to 5–12 min — a 60–83% reduction consistent with the linear flow-rate model. These results confirm the functional feasibility of the integrated sequence and provide a quantitative baseline for subsequent refinement.

B. Future Work

Future work should extend experimental validation for both subsystems through repeated trials using commercial kibble with different sizes, geometries, and compositions. Dosing accuracy should be characterized through extended repeated trials reporting mean error, standard deviation, and maximum error under varying hopper fill and kibble geometries; gravimetric closed-loop correction via load cell is the primary path to the ≤ 15% target. For the moistening stage, future work should quantify water uptake, final moisture content, and kibble fracture force as functions of exposure time and atomizer count, complemented by a water mass balance distinguishing food absorption from condensation and vapor losses. Drive torque should be measured directly under representative operating conditions to validate the sizing model of Eqs. (1)–(2). Extended multi-cycle tests should characterize mist distribution uniformity, condensate accumulation, and cleaning requirements. Any evaluation of clinical benefit for cats with mastication difficulties requires a separate protocol under appropriate veterinary and ethical oversight and falls outside the scope of the present work.

ACKNOWLEDGMENT

The authors acknowledge PUCP's Mechatronics Engineering Program for providing facilities and resources, and the Faculty of Sciences and Engineering for financial support through the Fondo Extraordinario de Apoyo a la Investigación para Estudiantes (FEAPE). The prototype was initially developed as part of the undergraduate course 1MTR52 (Mechatronic Design Project).

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